

Lab 8: Large-Sample Inference

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8.1 NBA Free Throws

8.1.1 Data

NBA free throw data from the 2022-2023 and 2023-2024 seasons is stored in `08_nba-free-throws.csv`. Each row is a player-team combination, and each contains the following variables:

- **Player:** the player's name
- **Team:** the team the player played for
- **G:** the number of games the player played
- **FT:** the number of free throws made per game
- **FTA:** the number of free throws attempted per game

Throughout this lab, treat free throws as a simplified Binomial model: for a given player-season, assume attempts are independent and that the player's free-throw probability p is constant over the period being studied.

8.1.2 Your Task

1. Modify the above dataset to include the following variables:
 - **Player:** the player's name
 - **FT_total:** total number of free throws made
 - **FTA_total:** total number of free throws attempted
 - **FT_percent:** free throw percentage across a season
2. Filter the dataset to include only players who shot at least 25 free throws.
3. For each player, plot $\hat{p} = \text{FT}\%$ (x axis) vs player name (y axis), overlaying the Wald and Agresti-Coull confidence intervals. What do you observe?

8.2 LLN Warm-Up

8.2.1 Your Task

1. Simulate Bernoulli free throws with $p = 0.75$.

- For one simulated sequence of length 1000, compute the running estimate

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

for $n = 1, \dots, 1000$, and plot $\hat{p}_n$ against n . Add a horizontal line at $p = 0.75$.

- Repeat the simulation 100 times and overlay the 100 paths of $\hat{p}_n$. What happens to the variability as n grows?

8.3 Simulation Study

We want to understand how Wald and Agresti-Coull confidence intervals perform across a range of values of p . To do this, we will simulate data from a binomial distribution with varying values of p and compute the confidence intervals for each.

For the Agresti-Coull interval, use the general formula

$$\begin{aligned}\tilde{n} &= n + z_{0.975}^2 \\ \tilde{p} &= \frac{S_n + z_{0.975}^2/2}{\tilde{n}} \\ \text{CI}_{AC} &= \tilde{p} \pm z_{0.975} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}\end{aligned}$$

with $z_{0.975} = 1.96$. This is approximately the familiar “add 2 successes and 2 failures” rule for a 95% interval. Equivalently, the general formula acts like adding $z_{0.975}^2/2$ artificial successes and $z_{0.975}^2/2$ artificial failures, which is approximately 2 and 2 at the 95% level.

8.3.1 Your Task

- Discretize the interval $[0, 1]$ into 1000 equally spaced points.
- For each value of p and each n in $\{10, 50, 100, 250, 500, 1000\}$, generate n free throws using the binomial distribution with parameter p .
- Compute the 95% Wald and Agresti-Coull confidence intervals for each value of p and n .
- Repeat steps 2 and 3 $M = 100$ times each.
- For each (n, p) pair, calculate in what percentage of the M simulations the Wald and Agresti-Coull confidence intervals contain the true parameter p . Plot this **coverage probability** vs p for each n and interval method (faceted).

8.4 Skittles Demo

You have a Skittles bag with n total Skittles and r red ones. Suppose the color of each Skittle is drawn independently from some distribution.

8.4.1 Your Task

1. Construct a 95% Wald confidence interval for the true probability that Mars, Inc. makes a Skittle red.
2. Construct a 95% Agresti-Coull confidence interval for that same probability.
3. Compare the two intervals. Which one seems more sensible if your observed red proportion is close to 0 or 1?